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INCH-POUND

MIL-PRF-32293 (AR)
13 June 2008

PERFORMANCE SPECIFICATION

CAP, BLASTING, ELECTRIC - M6

This specification is approved for use by the U.S. Army Armament Research, Development, and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the performance requirements and the verification provisions for the CAP, BLASTING: electric, M6.

1.2 Soldier module. The blasting cap is issued as a soldier module, consisting of a cap assembly (cap body with external lead wires), spool and individual protective cover.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not documents are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those specified in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-595/30045 – Brown, Lusterless

Comments, suggestions, or questions on this document should be addressed to (Commander ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000) or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-464 - Electromagnetic Environmental Effect Requirements for Systems
MIL-STD-650 - Explosive, Sampling, Inspection and Testing
MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those specified in the solicitation or contract.

ARDEC Drawings

8830954 Spool
12913874 Holder Assembly

(Copies of these drawings are available from US Army ARDEC, AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000, or online from Drawing-Request@pica.army.mil.)

US ARMY FIELD MANUALS

FM 3-34.214 - Explosives and Demolitions

(This Field Manual may be viewed and printed at http://www.apd.army.mil/, or ordered from Commandant, U.S. Army Engineer School, Ft. Leonard Wood, MO 65473-6650)

US ARMY DEVELOPMENTAL TEST COMMAND

ITOP 1-2-601 - Laboratory Vibration Schedules
ITOP 4-2-602 - Rough Handling Tests
ITOP 7-2-512* - Simulated Air Drop - Weapons and Individual Equipment

(Copies of these documents may be ordered from the US Army Developmental Test Command, ATTN: Publications, 314 Longs Corner Road, Aberdeen Proving Ground, MD 21005-5005, or online at http://www.dtc.army.mil/publications/topsindex.aspx. *ITOP 7-2-512 is available at: https://vdls.atc.army.mil/, with verification of authorization to receive and store classified information.)

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CODE OF FEDERAL REGULATIONS

49 CFR, parts Hazardous Material Regulations 100-199

(Copies of this document are available online at www.gpoaccess.gov/cfr/index.html or from the Superintendent of Documents, U.S. Government Printing Office, North Capitol & "H" Streets, N.W., Washington, DC 20402-0002.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein.

ASTM INTERNATIONAL

ASTM B 33 - Specification for Tinned Soft or Annealed Copper Wire for Electrical Purposes
ASTM B 749 - Standard Specification for Lead and Lead Alloy Strip, Sheet, and Plate Products
ASTM D 149 - Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies
ASTM D 2633 - Standard Test Methods for Thermoplastic Insulations and Jackets for Wire and Cable

(Copies of ASTM standards may be ordered online at http://www.astm.org/ or from the ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design verification. When specified, samples of the blasting cap shall be subjected to design verification in accordance (IAW) with 4.1.

3.2 First article. When specified in the contract or purchase order (see 6.2), samples shall be subjected to first article inspection in accordance with 4.2.

3.3 Conformance inspection. When specified (see 6.2), samples of the blasting cap shall be subjected to conformance inspection in accordance with 4.3.

3.4 Interface and interoperability requirements.

3.4.1 Blasting cap body.

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3.4.1.1 Blasting cap body material. The blasting cap body shall be a non-magnetic casing capable of protecting energetic material and sufficiently strong to allow insertion into designated demolition materials (see 6.7). An extruded aluminum cup with side-wall thickness within the range of 0.008 to 0.0110 inches has been found acceptable in the past.

3.4.1.2 Blasting cap body dimensions. Length of blasting cap body shall be not greater than 2.35 inches and not less than 2.29 inches. The Outside Diameter of the cap shall be not greater than 0.241 inches and not less than 0.237 inches. The minimum diameter shall not apply to crimps or grooves.

3.4.2 External lead wires (shunt critical). The blasting cap shall have two, individually insulated electrical lead wires with the following physical characteristics:

a. The lead wires diameter shall be 25.3 mils nominal (AWG22) conforming to ASTM B 33 standard and not less than 143 nor greater than 150 inches in length.

b. The wire insulation shall have a dielectric breakdown voltage of not less than 600 volts.

c. Insulation wire resistance shall be not less than 2.4×10^6 ohms.

d. Insulation shall be brown in color and be suitable for use in underwater demolition.

e. The external ends of the wire shall be stripped 2.25 + .25 inches and the stripped ends shall be shunted (critical) by twisting the wires together or with the provision of a shunting device.

f. Resistance of blasting cap, including lead wires, shall be not greater than 1.6 +/- 0.16 ohms.

3.4.3 Spool. The blasting cap shall be provided with a spool, for impact protection of the blasting cap and for winding of the external lead wires. The spool dimensions shall be as specified on 8830954. The lead wires shall be tightly and neatly coiled around the length of the outside of the spool in an untangled manner.

3.4.4 Individual protective cover. The cap and spool shall be encased in an individual protective cover capable of preventing initiation from electro-static discharge (ESD). A sealed Mylar “sleeve” has been successfully used in the past as the protective cover.

3.4.5 Holder interface. The soldier module shall fit into a prescribed holder assembly as specified on 12913874 to prevent explosive propagation when packed in the M19A1, Ammunition Can.

3.5 Operating requirements.

3.5.1 Explosive output. When functioned within the specified temperature range, the blasting cap shall have sufficient explosive output to perforate a lead disc of 0.1345 inch thickness or initiate the following military explosives: C-4 blocks, 15/40 lbs., Shaped Charges, and TNT blocks.

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3.5.2 Ampere initiation of single cap. The M6 Blasting Cap shall function high-order when an electric DC current of not greater than 0.45 Amperes is applied to the lead wires for not more than 50 milliseconds.

3.5.3 Milliwatt-second initiation of single cap. The M6 Blasting Cap shall function high-order when electrical energy of 16.3 milliwatt-seconds is applied to the lead wires for not less than 10 milliseconds and not more than 24 milliseconds.

3.5.4 Fire conditions for fifty (50) caps in series. Fifty (50) M6 Blasting Caps connected in series through the full lengths of their leads shall be initiated by the application of a current of not less than 1.50 and not greater than 1.51 Amperes for not less than 1.5 and not more than 1.6 milliseconds.

3.5.5 Lead wire pull resistance. Lead wires shall not be separated from the blasting cap when a pull force of 5 pounds (lbf) is exerted on each lead wire while the blasting cap is held stationary.

3.6 Support and ownership requirements.

3.6.1 Reliability. The blasting cap shall have 99.9 percent functional reliability rating with 95% confidence.

3.6.2 Human factor.

3.6.2.1 M6 Blasting Cap. The M6 blasting cap shall be capable of being safely, rapidly, and reliably operated by the soldier wearing Cold/Wet Weather Glove and the NBC Protective Gloves.

3.6.2.2 Deployment from Soldier Module. The Soldier Module shall have no sharp edges and shall fit into the standard Battle Dress Uniform (BDU) breast pockets.

3.6.3 Safety

3.6.3.1 No fire (critical). The M6 cap shall not be initiated by an electric current of 200 to 210 milliamperes for the duration of 5 seconds.

3.6.3.2 Personnel ESD. In the Soldier Module, the blasting cap shall not initiate when subjected to Personnel (human body) ESD environments.

3.6.3.3 Helicopter ESD. The M6 Blasting Cap in the Soldier Module and packaged IAW packaging provisions of the contract (see 6.1.1), shall not be initiated when subjected to Helicopter ESD environments of a discharge from a 1000 Pico Farads capacitor charged to 300,000 volts.

3.6.3.4 Hazards of electromagnetic radiation to ordnance (HERO). The M6 Blasting Cap in the Soldier Module shall not initiate from the external radio frequencies (RF) and energy

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levels as specified in MIL-STD-464 “External EME for deck operations on ships” and Test F-3, Appendix F of MIL-STD-331.

3.6.3.5 Cups Split or Cracked and Explosive Leakage (critical). The M6 Blasting Cap shall not exhibit any cracks or splits, nor show any explosive exudation.

3.6.4 Energetic material qualification. The energetic material shall be qualified and approved for intended purposes by the Army Qualification Authority (see 6.6).

3.6.5 Energetic materials compatibility. All energetic materials (explosives, propellants, and pyrotechnics) shall be compatible with all combinations of energetic and non-energetic materials in direct contact. In addition, they shall be compatible with all combinations of materials that have the potential for contact in the system's life-cycle.

3.6.6 Storage life. The M6 Blasting Cap in Soldier Module and packaged in accordance with the packaging provisions of the contract, shall remain safe and functional with no degradation in performance for not less than 20 years in protected storage and for not less than 2 years in unprotected storage in all climatic conditions.

3.6.7 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168. (see 6.2).

3.6.8 Workmanship. All components shall be free of chips, dirt, grit, or other foreign material. The cleaning methods used shall not damage any assembly or parts nor shall the assembly or parts be contaminated by the cleaning agents employed. There shall be no chemical or electrochemical corrosive effects from the manufacturing process.

3.6.9 Hazard classification. When packaged in accordance with the packaging provisions of the contract, the blasting caps shall have no more severe hazard classification than the following:

a. Multi-cap configurations: DOD Hazard Class/Div: 1.1
DOD Hazard Compatibility Group: B
DOT Hazard Class: 0030
UN Number: 0030
UN NOMENCLATURE: DETONATORS, ELECTRIC

b. Single cap configuration: DOD Hazard Class/Div: 1.4
DOD Hazard Compatibility Group: B
DOT Hazard Class: 0255
UN Number: 0255
UN NOMENCLATURE: DETONATORS, ELECTRIC

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3.7 Environmental requirements.

3.7.1 Operating temperature. The M6 Blasting Cap shall be fully functional in the temperature range from -40 ºF to +125 ºF.

3.7.2 Storage temperature. The M6 Blasting Cap in the Soldier Module shall remain functional at all required operating temperatures after exposure to storage temperatures from -65 °F to +165 °F.

3.7.3 Thermal shock. The M6 Blasting Cap in the Soldier Module shall not show evidence of material degradation, damage or failure, and its performance shall not be degraded after exposure to the three (3) cycles of thermal shock between temperature extremes of -60 ºF and +160 ºF.

3.7.4 High temperature and humidity. The M6 Blasting Cap in Soldier Module shall not show evidence of material degradation, damage or failure, and shall be operable after exposure to high temperature of 160 ºF + 5 ºF and relative humidity of 100 percent.

3.7.5 Solar (UV) radiation. The M6 Blasting Cap in the Soldier Module shall show no evidence of material degradation, damage or failure, and shall be operable after being exposed to solar radiation.

3.7.6 Fungus. The M6 Blasting Cap shall show no evidence of material degradation, damage or failure and shall be operable after exposure to insects, fungi and molds in both Soldier Module and unprotected cap configurations.

3.7.7 Salt/fog. The M6 Blasting Cap shall show no evidence of material degradation, damage or failure, and shall be operable after being subjected to the effects of salt/fog.

3.7.8 Dust. The M6 Blasting Cap shall show no evidence of material degradation, damage or failure, and shall be operable after exposure to the effects of blowing dust condition.

3.7.9 Sand. The M6 Blasting Cap shall show no evidence of material degradation, damage or failure, and shall be operable after exposure to the effects of blowing sand condition.

3.7.10 Waterproof. The M6 blasting cap shall function after submersion in water for not less than 4 hours at not less than 70 ^{o} F and not greater than 75 ^{o} F when the water is under a pressure of 50 pounds per square inches (gauge).

3.7.11 Lightning. The M6 Blasting Cap in the Soldier Module and packaged in accordance with the packaging provisions of the contract, shall not be degraded or damaged as a result of a near lightning strike in intermediate pack (M19A1 ammunition can), with and without the outer pack configuration (wire-bound box). See Section 6 for packaging specifics.

3.7.12 Five-foot drop. The M6 blasting cap shall remain safe and fully functional after exposure to a five-foot drop.

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3.7.13 Transportation handling/vibration. The M6 Blasting Cap in the Soldier Module and packaged in accordance with the packaging provisions of the contract, shall remain safe and fully functional after being subjected to military transportation handling/vibration conditions.

3.7.14 Air drop. The M6 Blasting Cap in the Soldier Module and packaged in accordance with the packaging provisions of the contract, shall be capable of low velocity airdrop without hazard to air crew or recovery personnel. The recovered M6 blasting caps shall have no physical damage, shall be safe to handle, and shall be fully functional.

4. VERIFICATION

TABLE I: Requirement/verification cross reference matrix

Method of Verification 1.- Demonstration 2.- Examination 3.- Test	**Class of Verification** A – Design Verification B – First Article C – Conformance			**Verification Level for Conformance** from MIL-STD-1916, Table I			
Section 3	**Verification Method**			**Verification Class / Level**			**Section 4**
Requirement	**1**	**2**	**3**	**A**	**B**	**C**	**Verification & Remark**
3.4.1.1 Blasting cap body material		X		X	X	IV	4.4.1
3.4.1.2 Blasting cap body dim.s		X		X	X	IV	4.4.1
3.4.2 External lead wires		X		X	X	IV	4.4.2
3.4.3 Spool		X		X	X	IV	4.4.3
3.4.4 Ind. protective cover		X		X	X	IV	4.4.4, 4.6.3.2
3.4.5 Holder interface		X		X	X	IV	4.4.4
3.5.1 Explosive output			X	X	X		4.5.1
3.5.2 Ampere initiation			X	X	X	10	4.5.1, 4.5.2, Note 1
3.5.3 Milliwatt-second initiation			X	X	X	10	4.5.1, 4.5.3, Note 1
3.5.4 50 caps in series			X	X	X	50	4.5.1, 4.5.4 Note 1
3.5.5 Lead wire pull resistance			X	X	X	70	4.5.1, 4.5.5, Note 2
3.6.1 Reliability			X	X			4.6.1
3.6.2.1 M6 blasting cap		X		X			4.6.2.1
3.6.2.2 Deployment from Soldier Module		X		X	X	IV	4.6.2.2
3.6.3.1 No fire			X	X	X	100%	4.6.3.1, Note 1
3.6.3.2 Personnel ESD			X	X			4.6.3.2
3.6.3.3 Helicopter ESD			X	X			4.6.3.3
3.6.3.4 HERO			X	X			4.6.3.4
3.6.3.5 Cups split or cracked		X		X	X	100%	4.6.3.5
3.6.4 Energetic material qualification		X	X	X			4.6.4
3.6.5 Energetic materials compatibility		X	X	X			4.6.5
3.6.6 Storage life			X	X			4.6.6
3.6.7 Ammunition lot numbering		X				II	4.6.7
3.6.8 Workmanship		X		X	X	II	4.6.8

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TABLE 1. Requirement/verification cross reference matrix – continued

Requirement	1	2	3	A	B	C	Verification & Remark
3.6.9 Hazard classification	X			X	X		4.6.9
3.7.1 Operating temperature			X	X	X		4.7.1
3.7.2 Storage temperature			X	X			4.7.2
3.7.3 Thermal shock			X	X			4.7.3
3.7.4 High temperature & humidity			X	X			4.7.4
3.7.5 Solar (UV) radiation			X	X			4.7.5
3.7.6 Fungus			X	X			4.7.6
3.7.7 Salt/fog			X	X			4.7.7
3.7.8 Dust			X	X			4.7.8
3.7.9 Sand			X	X			4.7.9
3.7.10 Waterproof			X	X	X	80	4.7.10
3.7.11 Lightning			X	X			4.7.11
3.7.12 Five-foot drop			X	X			4.7.12
3.7.13 Transportation handling			X	X			4.7.13
3.7.14 Air drop			X	X			4.7.14
Note 1. Sequential Test: No fire test shall be performed on 100% of blasting caps following any temperature storage test but prior to function test. Note 2. Test Order: Check pull resistance first, then the no fire test, then a method of initiation for output.							

4.1 Design verification. Design verification shall be performed on a sample of 3000 M6 blasting caps which shall be subjected to design verification in accordance with TABLE I and TABLE II. See TABLE IV for test order of temperature-conditioned function tests.

TABLE II. Design verification test sequence and quantity

Requirement	Quantity	Test Method
3.4.1.1	100	4.4.1 Use same Sample for 3.4.1 through 3.4.5
3.4.1.2	100	4.4.1 Use same Sample for 3.4.1 through 3.4.5
3.4.2	100	4.4.2 Use same Sample for 3.4.1 through 3.4.5
3.4.3	100	4.4.3 Use same Sample for 3.4.1 through 3.4.5
3.4.4	100	4.4.4 Use same Sample for 3.4.1 through 3.4.5
3.4.5	100	4.4.4 Use same Sample for 3.4.1 through 3.4.5
3.5.1	----	4.5.1
3.5.2	600	4.5.1, 4.5.2
3.5.3	500	4.5.1, 4.5.3
3.5.4	1900	4.5.1, 4.5.4
3.5.5	All Sample	4.5.1, 4.5.5
3.6.1	----	4.6.1 Perform following all function testing.
3.6.2.1	10	4.6.2.1
3.6.2.2	10	4.6.2.2
3.6.2.3	10	4.6.2.3
3.6.3.1	All Sample	4.6.3.1
3.6.3.2	40	4.6.3.2

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TABLE II. Design verification test sequence and quantity – continued

Requirement	Quantity	Test Method
3.6.3.3	40	4.6.3.3
3.6.3.4	20	4.6.3.4
3.6.3.5	All Sample	.6.3.5
3.6.4	----	4.6.4
3.6.5	----	4.6.5
3.6.6	40	4.6.6, 4.5.1
3.6.8	40	4.6.8
3.6.9	----	4.6.9
3.7.1	----	4.7.1 See initiation methods below.
3.7.2	----	4.7.2 See initiation methods below.
3.7.3	40	4.7.3, 4.5.1
3.7.4	40	4.7.4
3.7.5	40	4.7.5
3.7.6	40	4.7.6
3.7.7	40	4.7.7
3.7.8	40	4.7.8
3.7.9	40	4.7.9
3.7.10	----	4.8.10 See initiation methods below.
3.7.11	40	4.7.11
3.7.12	See Table IV	4.7.12
3.7.13	See Table IV	4.7.13
3.7.14	80	4.7.14

Ambient Function Test (3.7.1 and 4.7.1.b)
Total Quantity: 950 M6's (Sequential Test)
Conduct Lead Wires Pull Resistance Test (4.5.5) then
Conduct No Fire Test (4.6.3.1) then function test as follows
350 M6's (7 sets of 50 caps per 4.5.4)
300 M6's (per 4.5.1 & 4.5.2)
300 M6's (per 4.5.1 & 4.5.3)

Hot Storage Test (3.7.2 and 4.7.2a)
Total Quantity: 900 M6's
Perform environmental conditioning then
Conduct No Fire Test (4.6.3.1) then function test as follows
700 M6's (14 sets of 50 caps per 4.5.4)
100 M6's (per 4.5.1 and 4.5.2)
100 M6's (per 4.5.1 and 4.5.3)

Cold Storage Test (3.7.2 and 4.7.2.b)
Total Quantity: 900 M6's
Perform environmental conditioning then
Conduct No Fire Test (4.6.3.1) then function test as follows
700 M6's (12 sets of 50 caps per 4.5.4)
100 M6's (per 4.5.1 and 4.5.2)
100 M6's (per 4.5.1 and 4.5.3)

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TABLE II. Design verification test sequence and quantity – continued

Waterproof Test (4.7.10) Total Quantity: 250 M6's Conduct Waterproof test (4.7.10) then conduct function test as following 150 M6's (3 sets of 50 caps per 4.5.4) 100 M6's (per 4.5.2)

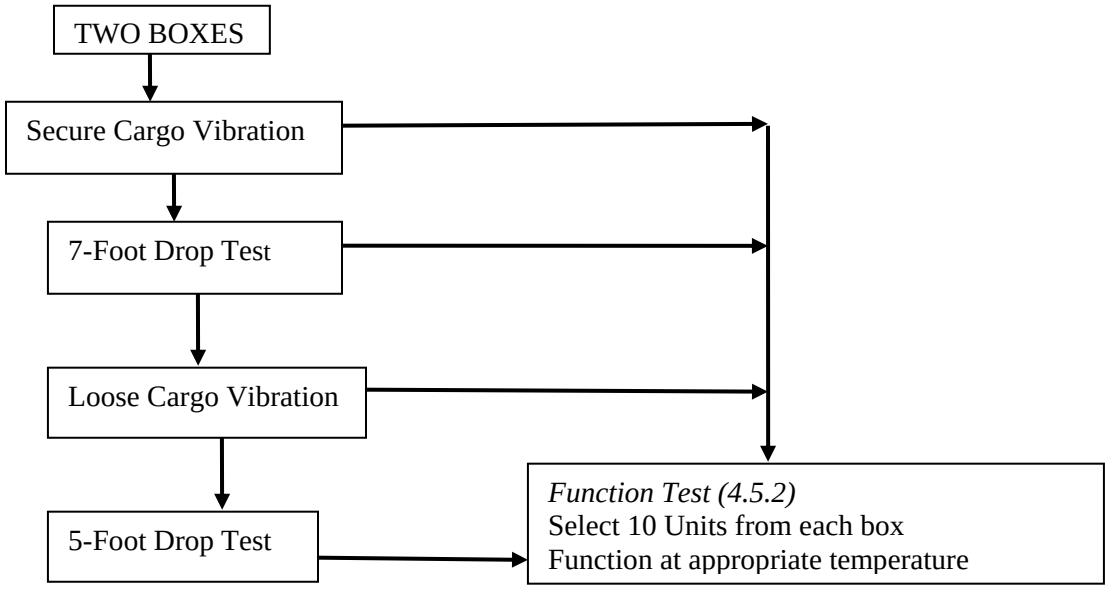
4.1.1 Design verification test rejection. If any test sample fails to comply with the design verification requirements (TABLES I & II), the design shall be rejected.

TABLE III. 5 Foot drop and Transportation handling/vibration test (see 4.7.12 and 4.7.13)

Total Qty: 6 Boxes (2 boxes at each Temperature)

Conduct Sequence of tests below at three temperatures:

1. Cold Storage temperature then function test at the cold operating temperature
2. Ambient Temperature
3. Hot Storage temperature then function test at the Hot operating temperature



4.2 First article inspection. When specified, a sample of the M6 blasting caps shall be subjected to first article verification test in accordance with TABLES I & III. See TABLE IV for test order of temperature-conditioned function tests.

TABLE IV. First article inspection

Requirement	Quantity	Test Method
3.4.1.1	50	4.4.1 Use same Sample for 3.4.1 through 3.4.5
3.4.1.2	50	4.4.1 Use same Sample for 3.4.1 through 3.4.5
3.4.2	50	4.4.2 Use same Sample for 3.4.1 through 3.4.5
3.4.3	50	4.4.3 Use same Sample for 3.4.1 through 3.4.5

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ABLE IV. First article inspection – continued

Requirement	Quantity	Test Method
3.4.4	50	4.4.4 Use same Sample for 3.4.1 through 3.4.5
3.4.5	50	4.4.4 Use same Sample for 3.4.1 through 3.4.5
3.5.1	----	4.5.1
3.5.2	100	4.5.2
3.5.3	40	4.5.3
3.5.4	750	4.5.4
3.5.5	All Sample	4.5.5
3.6.2.2	40	4.6.2.2
3.6.3.1	All Sample	4.6.3.1
3.6.3.5	All Sample	4.6.3.5
3.6.8	40	4.6.8
3.6.9	----	4.6.9
3.7.1	275	4.7.1 See below
3.7.2	550	4.7.2 See below
3.7.10	65	4.7.10
Ambient Function Test (3.7.1 and 4.7.1.b) Total Quantity: 275 M6's (Sequential Test) Conduct Lead Wires Pull Test (4.5.5) then Conduct No Fire Test (4.6.3.1) then function test as follows 200 M6's (4 sets of 50 caps per 4.5.4) 35 M6's (per 4.5.1 & 4.5.2) 40 M6's (per 4.5.1 & 4.5.3) Hot Storage Test (3.7.2 and 4.7.2.a) Total Quantity: 275 M6's Perform environmental conditioning then Conduct No Fire Test (4.6.3.1) then function test as follows 250 M6's (5 sets of 50 caps per 4.5.4) 25 M6's (per 4.5.1 & 4.5.2) Cold Storage Test (3.7.2 and 4.7.2.b) Total Quantity: 275 M6's Perform environmental conditioning then Conduct No Fire Test (4.6.3.1) then function test as follows 250 M6's (5 sets of 50 caps per 4.5.4) 25 M6's (per 4.5.1 & 4.5.2) Waterproof Test (4.7.10) Total Quantity: 65 M6's Conduct Waterproof test (4.7.10) then conduct function test as following 50 M6's (One set of 50 caps per 4.5.4) 15 M6's (per 4.5.2)		

4.2.1 First article rejection. If any test sample fails to comply with the first article requirements, the process/lot shall be rejected.

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4.3 Conformance verification. When specified, a sample of the M6 blasting caps shall be subjected to conformance verification in accordance with TABLE I and V.

TABLE V. Conformance verification.

Requirement	Quantity	Test Method
3.4.1.1	VL IV	4.4.1 Use same Sample for 3.4.1, 3.4.2, and 3.4.3
3.4.1.2	VL IV	4.4.1 Use same Sample for 3.4.1, 3.4.2, and 3.4.3
3.4.2	VL IV	4.4.2 Use same Sample for 3.4.1, 3.4.2, and 3.4.3
3.4.3	VL IV	4.4.3 Use same Sample for 3.4.1, 3.4.2, and 3.4.3
3.4.4	VL IV	4.4.4
3.4.5	VL IV	4.4.4
3.5.2	10	4.5.2
3.5.3	10	4.5.3
3.5.4	50	4.5.4
3.5.5	70	4.5.5
3.6.2.2	VL IV	4.6.2.2
3.6.3.1	100%	4.6.3.1
3.6.3.5	100%	4.6.3.5
3.6.7	VL II	4.6.7
3.6.8	VL II	4.6.8
3.7.10	80	4.7.10
Waterproof Test (4.7.10) Total Quantity: 80 M6's Conduct Waterproof test (4.7.10) then conduct function test as following 50 M6's (One set of 50 caps per 4.5.4) 30 M6's (per 4.5.2)		

4.3.1 Lot formation. Lot formation shall be in accordance with the lot formation and identification requirements listed in MIL-STD-1916.

4.3.2 Classification of characteristics. Conformance inspection shall consist of all the examination and tests listed in TABLE I. Three verifications, 4.6.3.1, 4.6.3.5, and 4.4.2 (shunt) are classified as critical characteristics. All other verifications are classified as major characteristics except for 4.6.7 and 4.6.8 which are classified as minor characteristic. For definition of Major and Minor characteristics, see MIL-STD-1916.

4.3.3 Lot Rejection. If any samples fail to comply with any other conformance inspection requirement, the lot shall be rejected.

4.4 Interface and interoperability verification.

4.4.1 Blasting cap body verification. The physical dimensions of the blasting cap body shall be measured using standard measuring equipment capable of measuring down to one thousandth of an inch. The non-magnetic characteristic shall be verified by examination.

4.4.2 External lead wires verification. The M6 blasting cap shall be inspected to ensure the lead wires are in compliance. Unspool lead wires from the spool and measure to verify that

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the maximum external dimension of the lead wires, excluding the shunt shall not exceed 150 inches and not be less than 143 inches in length, and that the wire insulation color is brown, IAW FED-STD-595/30045. Inspect the wire insulation for uniform and continuous coverage. Verify dielectric breakdown voltage on the insulation per ASTM D 149 and the insulation resistance per ASTM D 2633. Verify the M6 blasting cap resistance and continuity including lead wires with a current no more than 20 milliamperes. Combination of lead/bridge wire resistance shall be no more than 1.6 +/- 0.16 ohms.

4.4.3 Spool verification. Visually examine the spool to ensure it covers the entire M6 blasting cap and that the entire length of lead wires is tightly and neatly wound on the spool. Upon completion of the above visual inspection, uncoil the lead wires and remove the blasting cap from the spool. Verify by demonstration that the lead wires can be unwound in an untangled manner and the blasting cap can be easily removed from the spool. Failure in the above visual inspection and difficulty in unwinding the lead wires and removal of the blasting cap from the spool shall be considered as a failure.

4.4.4 Individual protective cover/holder interface. Visually inspect that blasting cap is enclosed in protective cover and demonstrate that it can be placed in Holder Assembly by hand and without difficulty.

4.5 Operating verification.

4.5.1 Explosive output test. The explosive output requirement shall be verified by either or both of the following verification methods.

4.5.1.1 Lead disc output test. The blasting caps shall be affixed in an upright position so that the base of the blasting cap is in contact with the center of a lead disc. The lead discs used in this test shall be not greater than 1.22 inches and not less than 1.21 inches in diameter, and not less than 0.1345 inches thick complying with ASTM B 749 for Lead Sheet material. The disc shall be centered over a 0.9375 (15/16) inches diameter hole in a test fixture. The blasting cap shall be initiated and observed for complete detonation and perforation of the lead discs.

4.5.1.2 Explosive initiation test. Verify by test that the M6 blasting cap functions high-order the following demolition items: C-4 blocks, 15 lbs/40 lbs Shaped Charges, and TNT blocks. The demolition items shall be conditioned to not greater than -35ºF and not less than -40ºF. Priming of the demolition items shall be in accordance with FM 3-34.214.

4.5.2 Ampere initiation. The lead wires of a single blasting cap shall be connected to a rheostat and ammeter, or other suitable means of controlling and measuring DC electric power. DC power no greater than 0.45 Amperes shall be applied, and the time from application of power to detonation of the blasting cap shall be measured using standard timing equipment capable of measuring to the nearest millisecond. Verify the blasting cap functions high-order by one of the demonstrations in 4.5.1.

4.5.3 Milliwatt-second initiation. The lead wires of a single blasting cap shall be connected to an electric energy meter, or other suitable means of controlling and measuring DC

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electric energy. DC energy no greater than 16.3 milliwatt-seconds shall be applied, and the time from application of power to detonation of the blasting cap shall be measured, using standard timing equipment capable of measuring to the nearest millisecond. Verify the blasting cap functions high-order by one of the demonstrations in 4.5.1.

4.5.4 Fifty (50) caps test. Fifty (50) M6 blasting caps shall be connected in series and initiated by applying current of 1.50 Amperes from power source. Suitable equipment to measure and control DC electric power shall be used, as well as equipment capable of measuring to the nearest millisecond the time from application to detonation of the blasting cap. Five (5) randomly selected blasting caps in the series connection shall have their explosive output verified in accordance with the procedures of 4.5.1.

4.5.5 Blasting cap lead wires pull resistance test. Affix the blasting cap horizontally to a fixture and apply a minimum 5 lbf pull force on each lead wire for both lead wires.

4.6 Support and ownership verification.

4.6.1 Reliability verification. The reliability shall be determined by analysis following the completion of all other testing.

4.6.2 Human factor verification.

4.6.2.1 M6 blasting cap. Verify by demonstration the M6 blasting cap priming operation can be performed by soldier wearing Mission Oriented Protective Posture (MOPP) IV/Cold wet gloves or Nuclear Biological Chemical (NBC) protective gloves.

4.6.2.2 M6 Blasting Caps in Soldier Module. Verify by examination that the M6 blasting caps in the Soldier Module do not contain any sharp edges which could cause injuries, and can be fitted into the standard Battle Dress Uniform (BDU) breast pockets.

4.6.3 Safety verification test.

4.6.3.1 No fire test. The lead wires of a single blasting cap shall be connected to a rheostat and ammeter, and DC power not greater than 210 milliamperes and not less than 200 milliamperes shall be applied. The time from application of power shall be measured using standard timing equipment capable of measuring to the nearest tenth of a second. No reaction of the cap shall be considered a success for this test.

4.6.3.2 Personnel ESD. The M6 Blasting Cap with lead wires in the shunted condition shall be subjected to ESD criteria of MIL-STD-331. Test F1. The human body ESD requirement for electrical initiator is a discharge from a 500 Pico Farads capacitor charged to 25,000 volts. No reaction of the cap shall be considered a success for this test.

4.6.3.3 Soldier Module ESD. The M6 Blasting Cap with lead wires shunted in the Soldier Module shall be subjected to Helicopter & Air ESD requirement of a discharge from a 1000 Pico

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Farads capacitor charged to 300,000 volts. No reaction of the cap shall be considered a success for this test.

4.6.3.4 HERO test. M6 blasting caps in the soldier module and with the lead wires shunted shall be subjected to the electromagnetic environmental conditions listed in Table IA of MIL-STD-464 and per Test F-3, appendix F of MIL-STD-331 without functioning.

4.6.3.5 Caps Split or Cracked and Explosive Leakage. Visually inspect the blasting caps for splits, cracks and evidence of explosive leakage.

4.6.4 Energetic material qualification. Objective evidence of Army Qualification Authority approval shall be examined (see 6.6).

4.6.5 Energetic material compatibility verification. Compatibility tests shall be conducted in accordance with Method 504.1.1 of MIL-STD-650 or Method 408.1.1 of MIL-STD-286. Compatibility is demonstrated when results reflect non-detectable reactivity.

4.6.6 Storage life verification. Verify the M6 blasting caps meet the storage life requirement by conducting the MIL-STD-331 test C1 in soldier module configuration. At the completion of test, blasting cap assemblies shall be function tested per 4.5.2 – 4.5.4.

4.6.7 Ammunition lot numbering verification. Verify by examination that an ammunition lot number has been assigned to each lot of blasting caps as specified in MIL-STD-1168.

4.6.8 Workmanship verification. Verify the workmanship on the production of the M6 blasting cap by visual inspection.

4.6.9 Final hazard classification verification. Verify by examination that the M6 blasting cap is in compliance with the FHC requirements specified at paragraph 3.7. FHC requirements shall be in accordance with Code of Federal Regulation 49 CFR. Parts 100-199.

4.7 Environmental verification.

4.7.1 Operating temperature verification test.

a. Hot temperature. Condition the M6 blasting cap at not less than 120ºF and not greater than 125ºF for a minimum of 24 hours. Perform functional test in accordance with procedure in 4.5.2 – 4.5.4 within five (5) minutes after the samples have been removed from the conditioning chamber.

b. Ambient temperature. Condition the M6 blasting cap at not less than 75ºF and not greater than 85ºF for a minimum of 24 hours. Perform functional test in accordance with procedure in 4.5.2 – 4.5.4 within five (5) minutes after the samples have been removed from the conditioning chamber.

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c. Cold temperature. Condition the M6 blasting cap at not greater than -35ºF and not less than -40ºF for a minimum of 24 hours. Perform functional test in accordance with procedure in 4.5.2 – 4.5.4 within (5) minutes after the samples have been removed from the conditioning chamber.

4.7.2 Storage temperature verification test. The blasting cap, when packaged in accordance with the packaging provisions of the contract, shall be subjected to the following storage temperatures.

a. Hot temperature. Condition the M6 blasting cap at not less than 160ºF and not greater than 165ºF for a minimum of 24 hours. Then adjust the temperature to the minimum operating temperature of not less than 120ºF and not greater than 125ºF and let stabilize for a maximum of 4 hours. Perform function test in accordance with procedure in 4.5.2 – 4.5.4 within (5) minutes after the samples have been removed from the conditioning chamber.

b. Cold temperature. Condition the M6 blasting cap at not greater than -60ºF and not less than -65ºF for a minimum of 24 hours. Then adjust the temperature to the maximum operating temperature of not greater than -35ºF and not less than -40ºF and let stabilize for a maximum of 4 hours. Perform function test in accordance with procedure in 4.5.2 – 4.5.4 within (5) minutes after the samples have been removed from the conditioning chamber.

4.7.3 Thermal shock test. Perform the following test on the M6 blasting cap in the packaged configuration for a minimum of three cycles between the temperature extremes of paragraph 3.7.3 in accordance with MIL-STD-810, Method 503.4, Procedure II. The minimum duration of the low and the high temperatures is one (1) hour. Upon completion of the test, visually inspected caps for signs of damages that have negative impact on physical form and fit requirements in section 3 of this specification. Any M6 blasting caps sustaining such damages shall be considered as a failure. Following the inspection, conduct the function test in accordance with procedure in 4.5.2 – 4.5.4.

4.7.4 High temperature and humidity test. Perform the humidity test IAW MIL-STD-331, Test C1 (Humidity) on the M6 blasting cap. Upon completion of the test, the samples shall be visually inspected for signs of damages that have negative impact on physical form and fit requirements in section 3 of this spec. Any M6 blasting caps sustaining such damages shall be considered as a failure. Following the inspection, conduct the function test IAW procedure in 4.5.2 – 4.5.4.

4.7.5 Sunlight/UV test. Perform the solar radiation test IAW MIL-STD-810, Method 505.3, procedure II for ten 24-hour cycles on the M6 blasting cap. Upon completion of this test, inspect the test samples for signs of obvious damages per MIL-STD-810, Method 505.3, Procedure II, Paragraph I-4.1. Any M6 blasting caps sustaining such damages shall be considered as a failure. Following the inspection, conduct the function test IAW procedure in 4.5.2 – 4.5.4.

4.7.6 Fungus test. Perform the test on the M6 blasting cap in accordance with MIL-STD-810, Method 508.4. Procedure II. Inspection shall be in accordance with MIL-STD-810, Method

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508.5. Following the inspection, function tests shall be conducted for remaining test samples IAW procedure in 4.5.2 – 4.5.4.

4.7.7 Salt fog test. Perform a ninety-six (96) hour, i.e. 4-24 hour cycles, salt spray of a 5% concentration (fog) test on the M6 blasting cap in accordance with MIL-STD-810, Method 509.4. Upon completion of this test procedure, inspect the M6 Blasting Cap for evidence of surface degradation such as flaking, pitting, blistering or separation of finish. Following the inspection, conduct the function test IAW procedure in 4.5.2 – 4.5.4.

4.7.8 Dust test. Perform the blowing dust test on the M6 blasting cap per MIL-STD-810, Method 510.4, Procedure I. Following the test perform a visual inspection, then conduct the function test in accordance with procedure in 4.5.2 – 4.5.4.

4.7.9 Sand test. The M6 blasting cap assembly shall be subjected to the blowing sand test in accordance with MIL-STD-810, Method 510.4, Procedure II. Following the test perform a visual inspection, then conduct the function test in accordance with procedure in 4.5.2 – 4.5.4.

4.7.10 Waterproof test. The M6 blasting cap shall be submersed in water at a depth pressure of 50 plus 0.50 pounds per square inch (gauge) for not less than 4 hours at 70 degrees plus or minus 5 degrees Fahrenheit (F). After the waterproofness test, conduct the no fire test per 4.7.2.1 then conduct the function test IAW procedure in 4.5.2 – 4.5.3 within 30 minutes.

4.7.11 Lightning effect test. The M6 blasting caps packaged in the intermediate package (M19A1 or M2A1 ammunition boxes) and in the tactical configuration shall be exposed to near lightning strike at 10 meters, as defined in MIL-STD-464 with the following characteristics: 200,000 ampere peak, time to peak 1.5 microseconds, time to half current 50 microseconds, 400 amperes continuing current last 400 milliseconds. Detonation shall constitute failure of the test.

4.7.12 Five-foot drop test. Perform the drop test IAW MIL-STD-331, TEST A4.1 on the bare (no packaging) M6 blasting caps. Upon completion of the test, visually inspect for signs of damages that have negative impact on physical form and fit requirements. Any M6 blasting caps sustaining such damage shall be considered as a failure. Following the inspection, conduct the function test IAW procedure in 4.5.2 – 4.5.4 and TABLE III.

4.7.13 Transportation handling/vibration test.

a. Perform the transportation handling test as specified in ITOP 4-2-602 on the packaged M6 blasting caps. Upon completion of the test, the samples shall be visually inspected for signs of damage. Any M6 blasting caps sustaining such damage shall be considered as a failure. Remaining test samples shall be function tested to determine the functionality of the item.

b. Perform the transportation vibration test as specified in ITOP 1-2-601 on the packaged M6 blasting caps. Upon completion of the test, visually inspect for signs of damages that have negative impact on physical form and fit requirements in section 3 of this spec. Any M6 blasting caps sustaining such damages shall be consider as a failure. Remaining test samples shall be function tested IAW 4.5.2 – 4.5.4.

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4.7.14 Air-drop test. Perform the simulated air-drop on the M6 Blasting Cap in the packaged configuration in accordance with the packaging provisions of the contract (see 6.1.1) as specified in ITOP 7-2-512. M6 blasting caps shall be tested in the fully packaged configuration (unit pack, intermediate pack and exterior pack). The tested samples shall be examined for visual damage. The undamaged and minimal damaged test samples shall be function tested per 4.5.2 – 4.5.4.

5. **PACKAGING**

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. **NOTES**

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The blasting cap covered by this specification is military unique (size of the cap well and higher output compared with commercial cap) and will be used only by military personnel for initiating all past and current fielded and developmental military explosives.

6.1.1 M6 deployment. The M6 is typically deployed in one of three protective packaging options (configurations) depending on the user unit/mission, as follows:

Option A contains 10 each M6 Blasting Caps, and is issued as 18 modules for a total of 180 blasting caps. It is intended for transportation of blasting caps between Ammunition Plants.

Option B contains 96 M6 Blasting Caps, and is issued as a wire-bound box containing four M19A1 ammunition cans, each with 24 blasting caps. It is intended for transportation of blasting caps between Ammunition Plants.

Option C contains M6 Blasting Cap in the Soldier Module. It is issued as a wire-bound box containing four M19A1 ammunition cans, each with ten Soldier Modules. It is intended for field deployment.

6.2 Acquisition requirements. The following is a list of data and acquisition documents that are not specified within this document but are necessary to award the contract. These data should be included in a procurement package:

a. Title, number and date of this specification and all documents referenced in Section 2, as well as documents that provide necessary information that are listed in Section 6.

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b. Requirements for submission of Design Verification test sample.
c. Requirements for submission of first article test sample.
d. Requirements for submission of inspection equipment designs.
e. Applicable National Stock Number (NSN).
f. Certificate of conformance for each lot or shipment of product.
g. Provisions for Government furnished material.
h. Packaging requirements (see 5.1). Special packaging requirements apply to this material as described on drawing numbers 3139685, 12913870, and 12913871. Contact AMSRD-AAR-AIL-P, Picatinny, NJ.
i. Requirement for submission of Item hazard classification.
j. Information needed to satisfy the requirements for qualification of energetic materials by the Army Service Qualification Authority, see 6.7. This information can be obtained in the desktop guide titled Explosive Hazard Classification Requirements for Performance Specifications that is available from U.S. Army ARDEC, ATTN: System Safety Office, AMSRD-AAR-QES, bldg 62, Picatinny Arsenal, NJ 07806-5000.

6.3 Acceptance inspection equipment (AIE). Provisions concerning the AIE and test facilities that will be used to verify the requirements of this specification should be specified in the contract.

6.4 Explosive output. An output charge of 12.4 to 14.5 grains (813 to 943 milligrams) of RDX, Type A or B, class C, military explosive as specified in MIL-DTL-398 consolidated into an aluminum alloy ASTM B209 5052D, T0 casing in four increments at pressure between 5000 and 6000 lb per square inch has been qualified to initiate all past and current fielded military explosives such as C4, TNT, HMX, LX14, PETN, A5, Tetryl, N5, RDX and all PAXs at the specified temperature requirements.

6.5 Packaging options. The Joint Hazard Classification System has assigned the following DOT Explosive Registration Numbers for the M6 blasting cap.

Option A: EX Number: 8710615
NSN 1375-01-192-9174
Packaging defined in MIL-C-45468

Option B: EX Number: 8801434
NSN 1375-00-283-9442
Packaging defined by drawing 3139685

Option C: EX Number: 9203515
NSN 1375-01-316-1229
Packaging defined by drawing 12913870

6.6 Energetic material qualification information. Qualification of energetic materials by the Army Service Qualification Authority requires the demonstration that energetic materials meet the requirements of STANAG 4170. This task should be addressed in the statement of work in the contract. Points of Contact for the Army Service Qualification Authority can be

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obtained from U.S. Army ARDEC, ATTN: Reliability Management Office, AMSRD-AAR-QES, bldg 62, Picatinny Arsenal, NJ 07806-5000.

6.7 Demolition materials. Designated demolition materials are C-4 blocks, TNT blocks, 15 and 40 lb. Shaped Charges.

6.8 Subject term (key word) listing.

Initiator
Energetic

Custodian:
Army-AR

Preparing Activity:
Army-AR
(Project: 1375-2008-006)

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